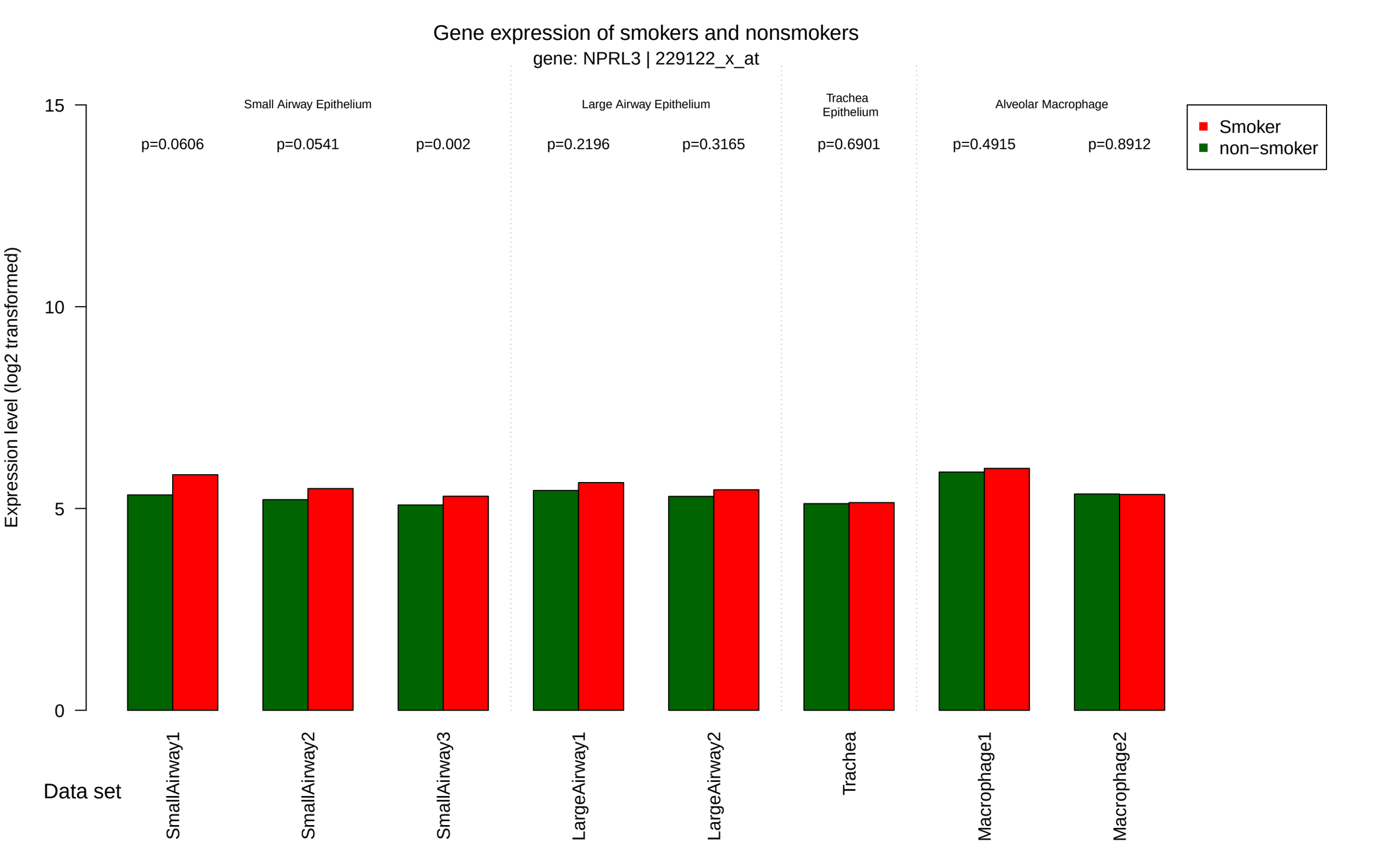
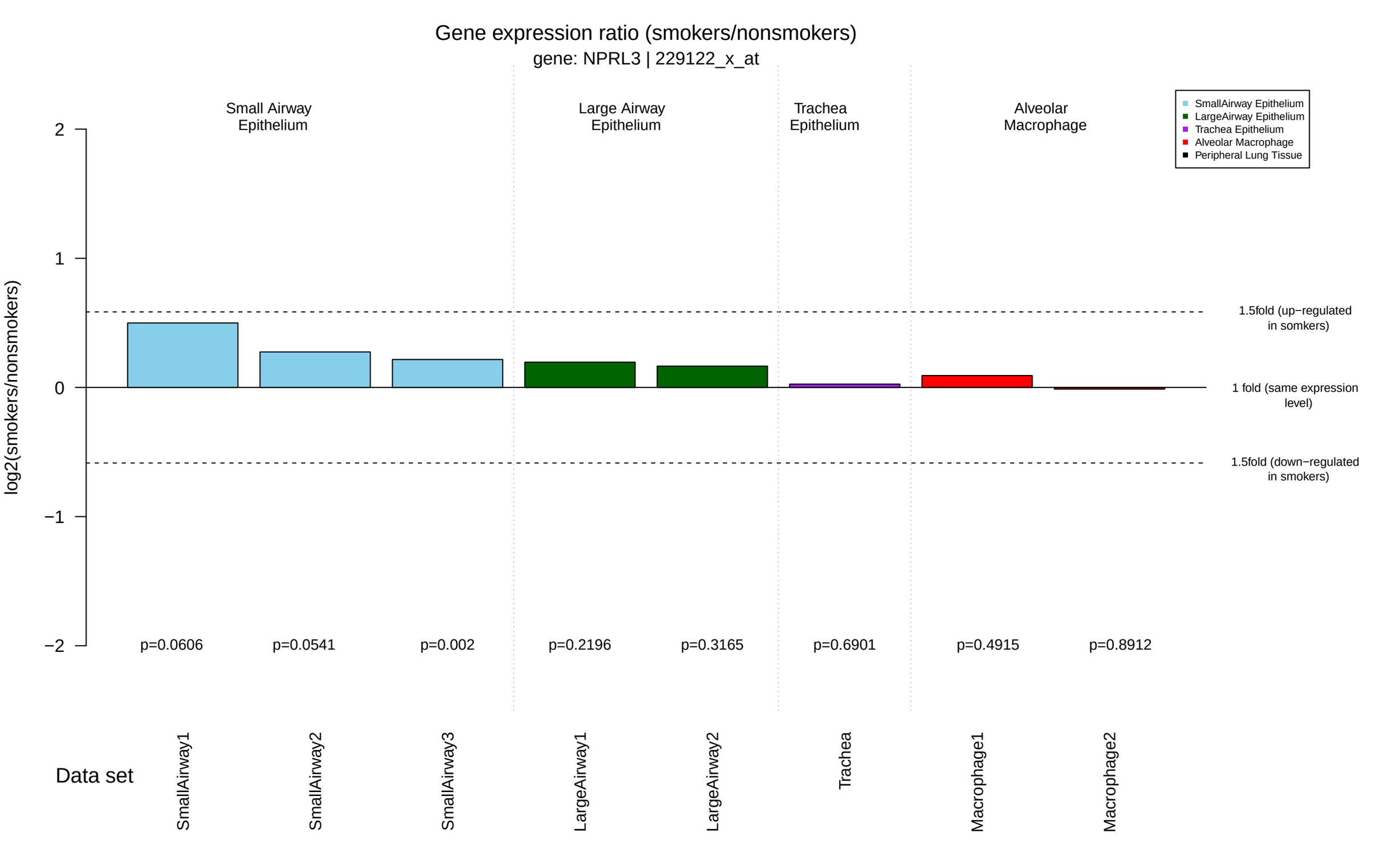
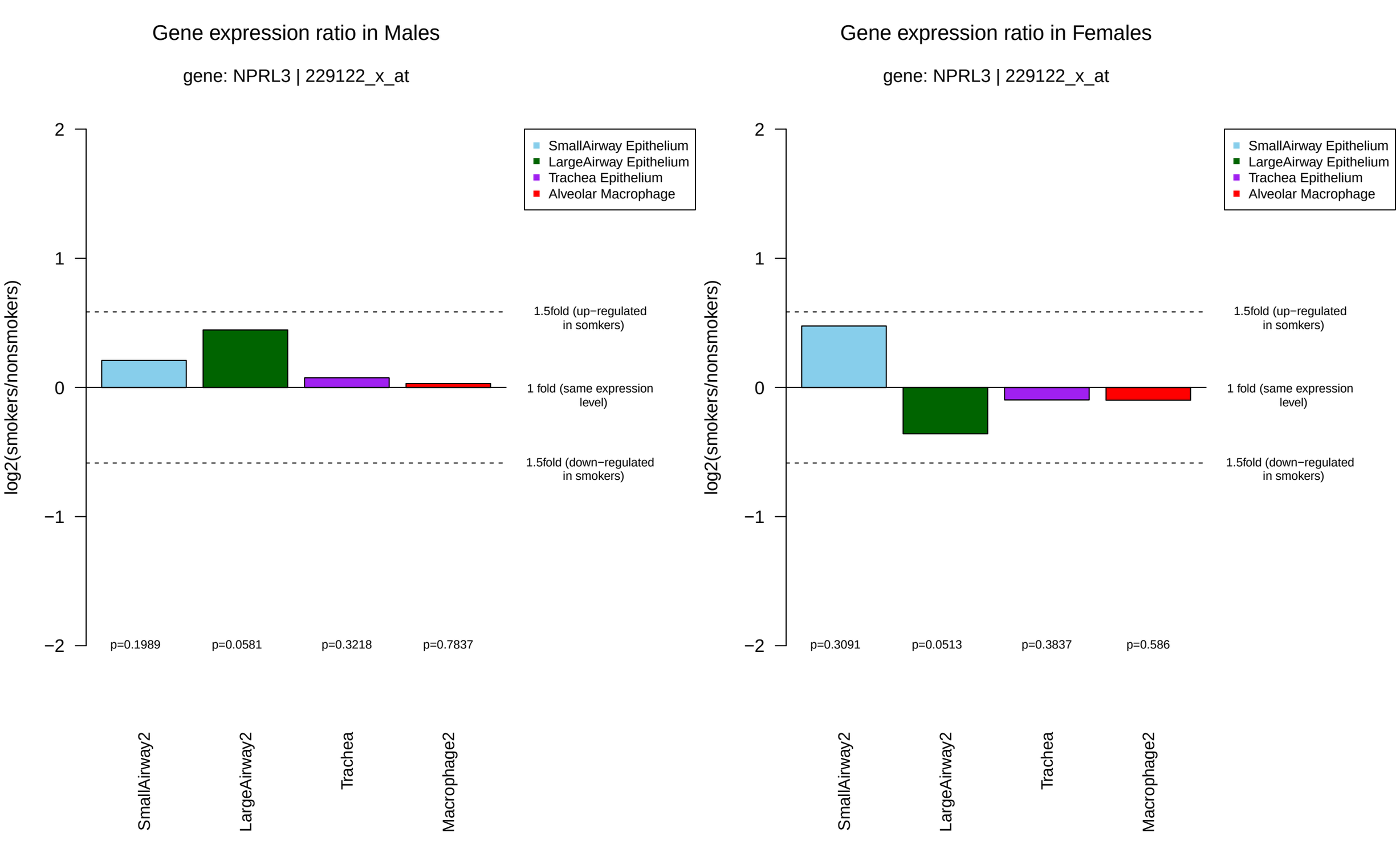
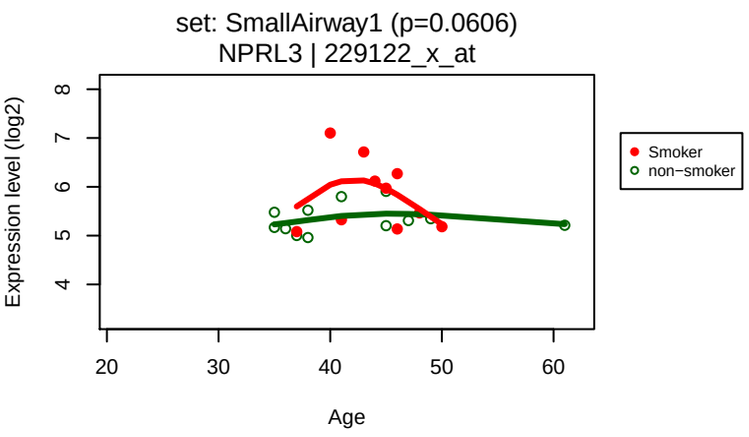
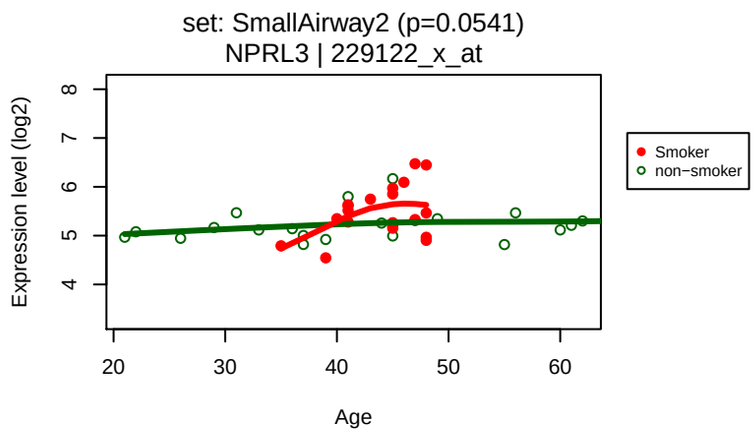
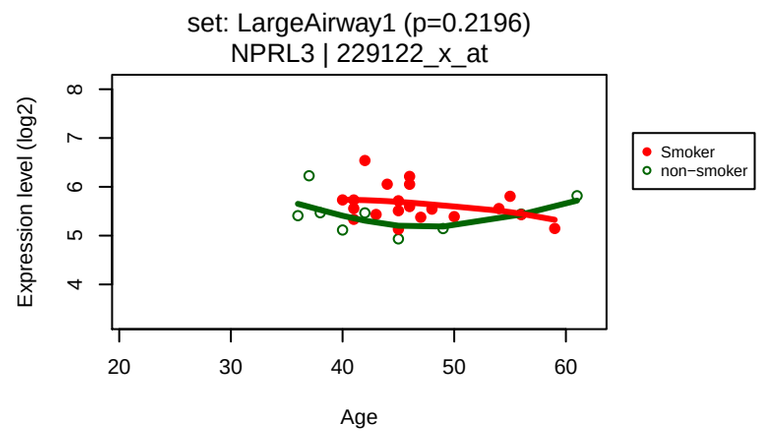

Gene expression of smokers and nonsmokers
gene: NPRL3 | 229122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0606
p=0.0541
p=0.002
p=0.2196
p=0.3165
p=0.6901
p=0.4915
p=0.8912
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

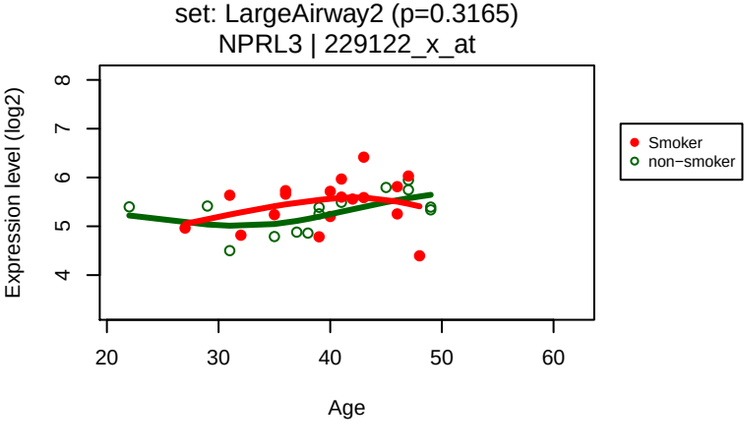

Gene expression ratio (smokers/nonsmokers)
gene: NPRL3 | 229122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0606
p=0.0541
p=0.002
p=0.2196
p=0.3165
p=0.6901
p=0.4915
p=0.8912
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

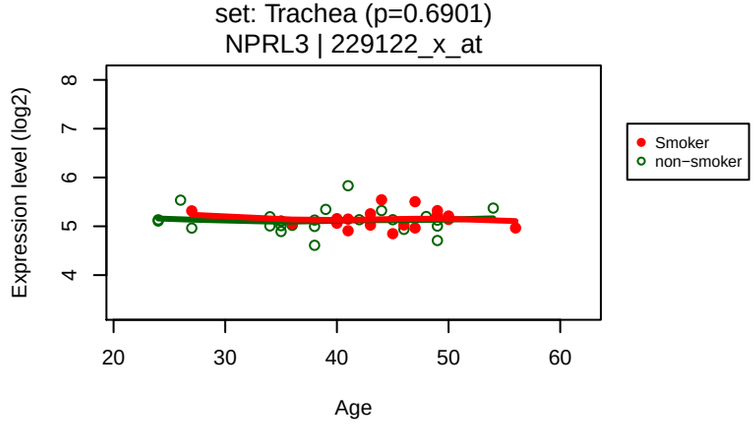

Gene expression ratio in Males
gene: NPRL3 | 229122_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1989
p=0.0581
p=0.3218
p=0.7837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPRL3 | 229122_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3091
p=0.0513
p=0.3837
p=0.586
SmallAirway2
LargeAirway2
Trachea
Macrophage2

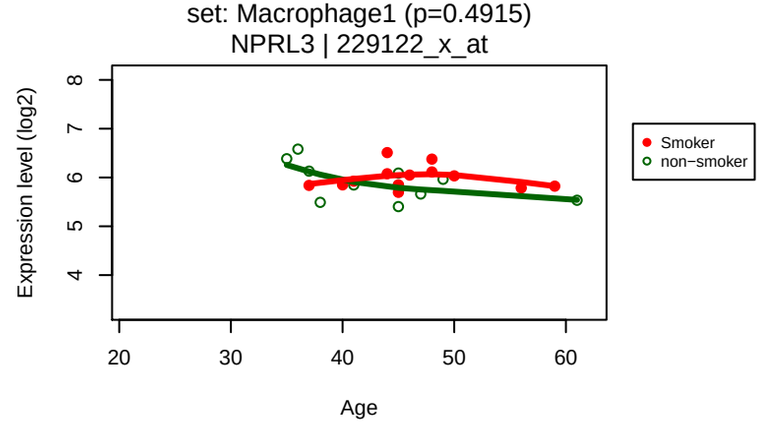

set: SmallAirway1 (p=0.0606)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker

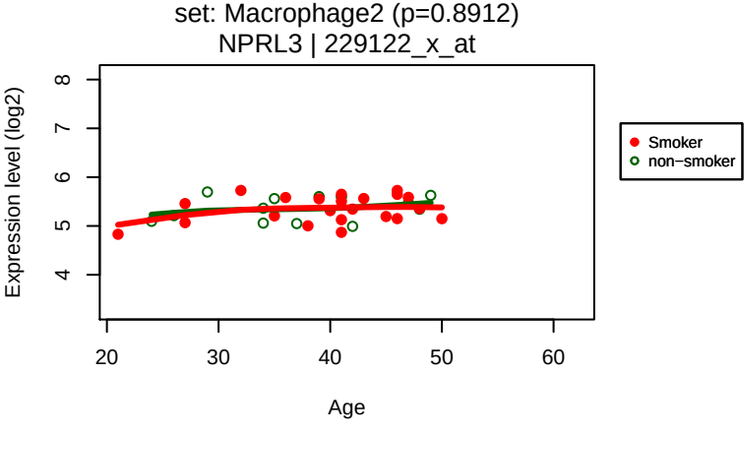

set: SmallAirway2 (p=0.0541)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker

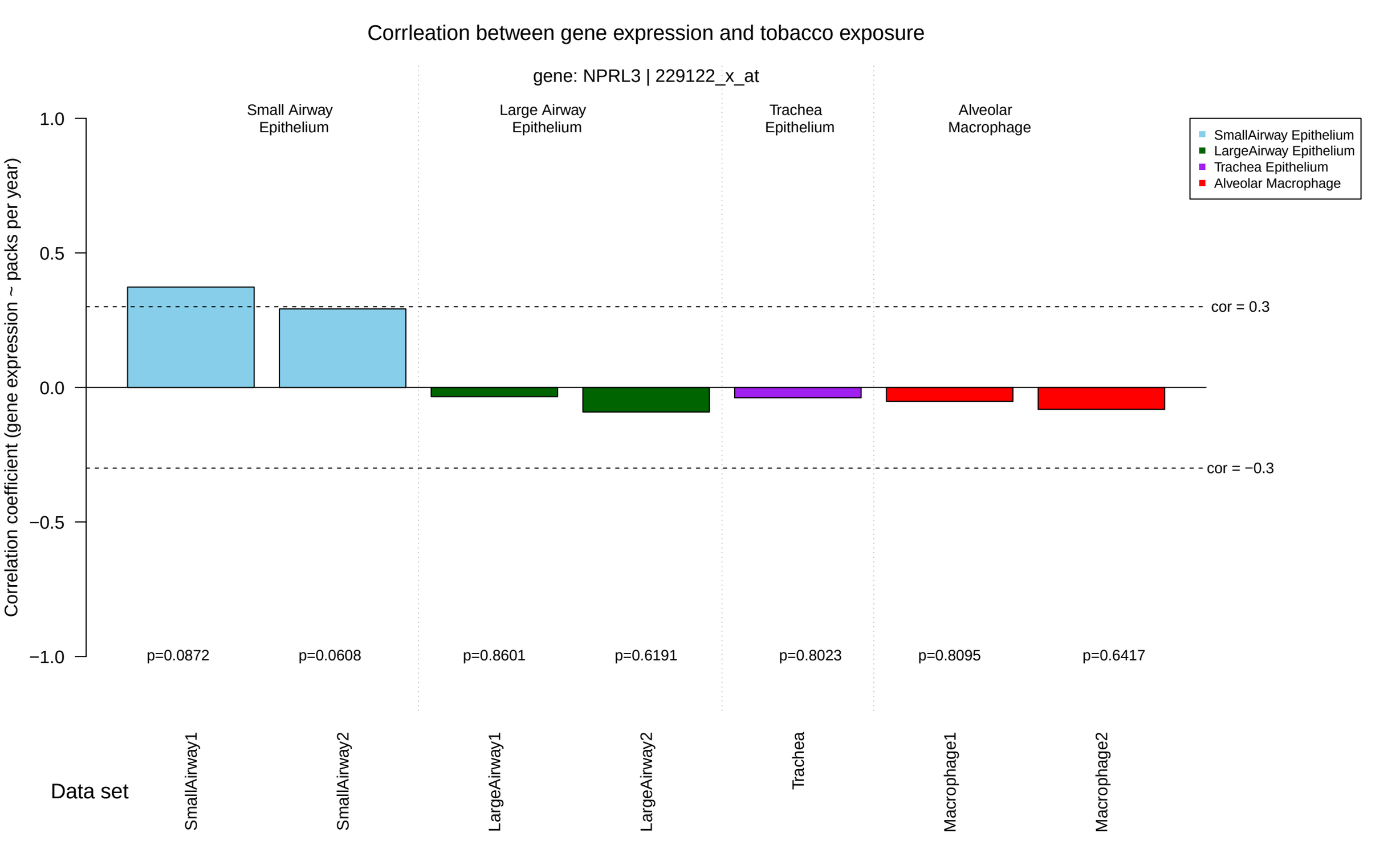

set: LargeAirway1 (p=0.2196)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3165)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6901)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4915)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8912)
NPRL3 | 229122_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPRL3 | 229122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0872
p=0.0608
p=0.8601
p=0.6191
p=0.8023
p=0.8095
p=0.6417
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2